Assignment 1: Kinematics

Open Duck Mini Robot

ROSE5710: Fundamental Robotics and Systems
First Term 2026–2027

1 Task overview

The task is to calculate the position and orientation of the robot's left foot, right foot and head using forward kinematics (FK), and to find joint angles for target poses using inverse kinematics (IK). Each leg has five joints; the head–neck chain has four.
For FK, joint angles are the input and site poses are the output. The image shows the supplied MuJoCo model with every joint angle set to zero, matching the D-H diagrams and zero-pose check.

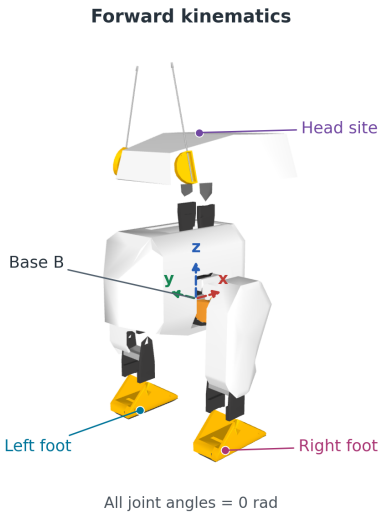


Figure 1: FK example: zero joint angles and the resulting robot pose.

2 Forward kinematics

2.1 Coordinate convention and calculation

All poses are expressed in the base frame; distances are in metres and angles in radians. The transform ${}^{i-1}\mathbf{T}_i$ maps coordinates from frame i to frame $i-1$.

For each D-H row, the transform is

$$\mathbf{A}_i = \mathbf{R}_x(\alpha_{i-1})\mathbf{T}_x(a_{i-1})\mathbf{R}_z(\theta_i)\mathbf{T}_z(d_i). \tag{1}$$

Writing $c_\theta = \cos\theta_i$, $s_\theta = \sin\theta_i$, $c_\alpha = \cos\alpha_{i-1}$ and $s_\alpha = \sin\alpha_{i-1}$ gives

$$\mathbf{A}_i = \begin{bmatrix} c_\theta & -s_\theta & 0 & a_{i-1} \\ c_\alpha s_\theta & c_\alpha c_\theta & -s_\alpha & -d_i s_\alpha \\ s_\alpha s_\theta & s_\alpha c_\theta & c_\alpha & d_i c_\alpha \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Joint signs and zero-angle offsets are included through $\theta_i = s_i q_i + \theta_i^0$. Starting from the base-to-frame-zero transform, the code multiplies the rows in order:

$${}^B\mathbf{T}_S(\mathbf{q}) = {}^B\mathbf{T}_0 \prod_{i=1}^{n+1} \mathbf{A}_i.$$

The final row places the site and has no joint variable. Code rows use $[\alpha, a, \theta^0, d]$; the tables use $[\alpha, a, d, \theta]$. The model is derived analytically and evaluated numerically.

2.2 Left leg

The hip-yaw offset defines the shared leg constants

$$r = \sqrt{0.019^2 + 0.035^2} = 0.0398246155\ \text{m}, \qquad \beta = \text{atan2}(19, 35) = 0.4973427163\ \text{rad}.$$

The initial transform has zero translation. The joint order is hip yaw, hip roll, hip pitch, knee and ankle:

$$^{B}\mathbf{T}_{0,L} = \mathbf{R}_z(\pi/2 + \beta), \qquad \mathbf{s}_L = [1, -1, -1, -1, -1]^{\mathsf{T}}.$$

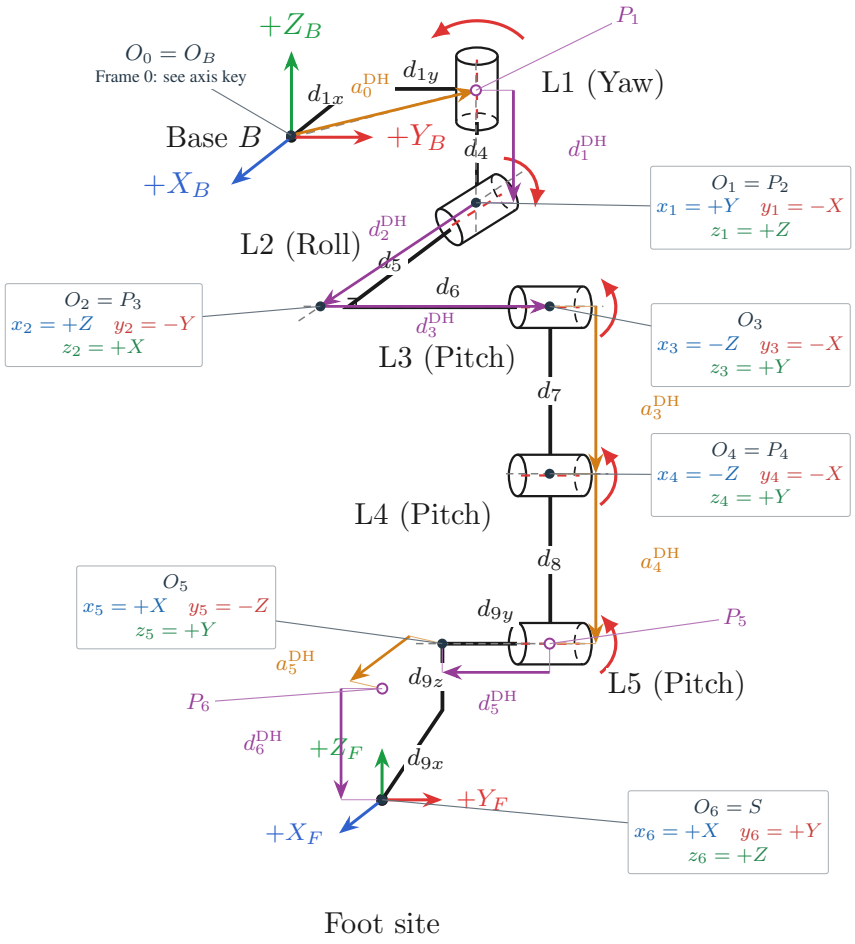


Figure 2: Left leg: annotated assignment Fig. 2. O_i: D-H origins; P_i: intermediate points. Orange: a_{i-1}; purple: d_i. Boxes give positive XYZ directions in B.

Table 1: Left leg: D–H parameters (m, rad) and zero-pose axes in frame B.

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	r	-0.0000741	$q_1 - \beta$
2	$\pi/2$	0	-0.01606	$-q_2 + \pi/2$
3	$\pi/2$	0	0.07415	$-q_3 + \pi$
4	0	0.07865	0	$-q_4$
5	0	0.07865	-0.01975	$-q_5 - \pi/2$
6*	$\pi/2$	0.0005	-0.036225	0

* Fixed foot site, not a sixth joint. Roll-axis height: $45.9409 - 46.015 = -0.0741$ mm.

i	$+x_i$	$+y_i$	$+z_i$
0	$-sX + cY$	$-cX - sY$	$+Z$
1	$+Y$	$-X$	$+Z$
2	$+Z$	$-Y$	$+X$
3	$-Z$	$-X$	$+Y$
4	$-Z$	$-X$	$+Y$
5	$+X$	$-Z$	$+Y$
6*	$+X$	$+Y$	$+Z$

X, Y, Z are base-frame unit axes. $s = \sin\beta$, $c = \cos\beta$.

2.3 Right leg

The joint order and constants r, β are the same as for the left leg. The base offset is mirrored, and the hip-pitch sign changes:

$$ {}^{B}\mathbf{T}_{0,R} = \mathbf{R}_z(-\pi/2 - \beta), \qquad \mathbf{s}_R = [1, -1, 1, -1, -1]^{\mathsf{T}}. $$

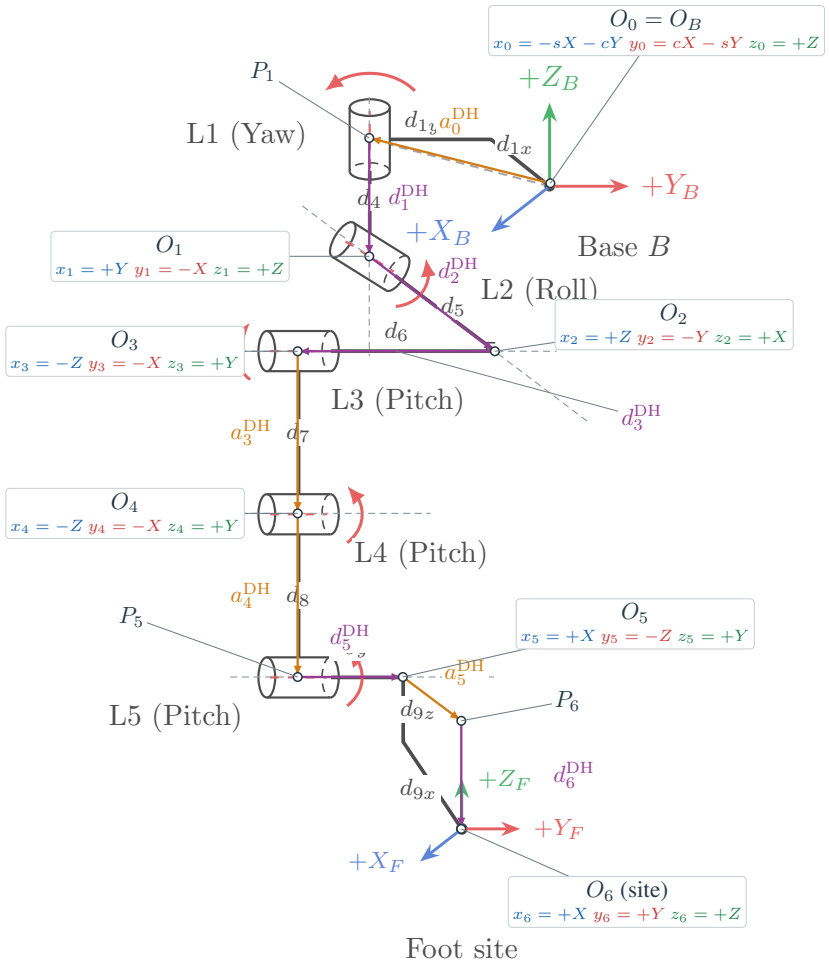


Figure 3: Right leg: annotated assignment Fig. 3. O_i: D-H origins; P_i: intermediate points. Orange: a_{i-1}; purple: d_i. Boxes give positive XYZ directions in B.

Table 2: Right leg: D–H parameters (m, rad) and zero-pose axes in frame B.

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	r	-0.0000741	$q_1 + \pi + \beta$
2	$\pi/2$	0	-0.01606	$-q_2 + \pi/2$
3	$\pi/2$	0	-0.07415	$q_3 + \pi$
4	0	0.07865	0	$-q_4$
5	0	0.07865	0.01735	$-q_5 - \pi/2$
6*	$\pi/2$	0.0005	-0.036225	0

* Fixed foot site. The combined lateral pitch-axis offset is 17.35 mm on the right (19.75 mm on the left).

i	$+x_i$	$+y_i$	$+z_i$
0	$-sX - cY$	$cX - sY$	$+Z$
1	$+Y$	$-X$	$+Z$
2	$+Z$	$-Y$	$+X$
3	$-Z$	$-X$	$+Y$
4	$-Z$	$-X$	$+Y$
5	$+X$	$-Z$	$+Y$
6*	$+X$	$+Y$	$+Z$

X, Y, Z are base-frame unit axes. $s = \sin\beta$, $c = \cos\beta$.

2.4 Head and neck

The initial transform includes the base-to-neck offset:

$$ {}^{B}\mathbf{T}_{0,H} = \begin{bmatrix} \mathbf{I}_3 & \mathbf{p}_c \\ \mathbf{0}^{\mathsf{T}} & 1 \end{bmatrix}, \qquad \mathbf{p}_c = \begin{bmatrix} 0.001 \\ 0.01915 \\ 0.0900009 \end{bmatrix} \text{m}. $$

Joint order is neck pitch, head pitch, head yaw and head roll. The zero-pose axes are $-Y, -Y, +Z, -X$, matching MuJoCo, so $\mathbf{s}_H = [1, 1, 1, 1]^{\mathsf{T}}$.

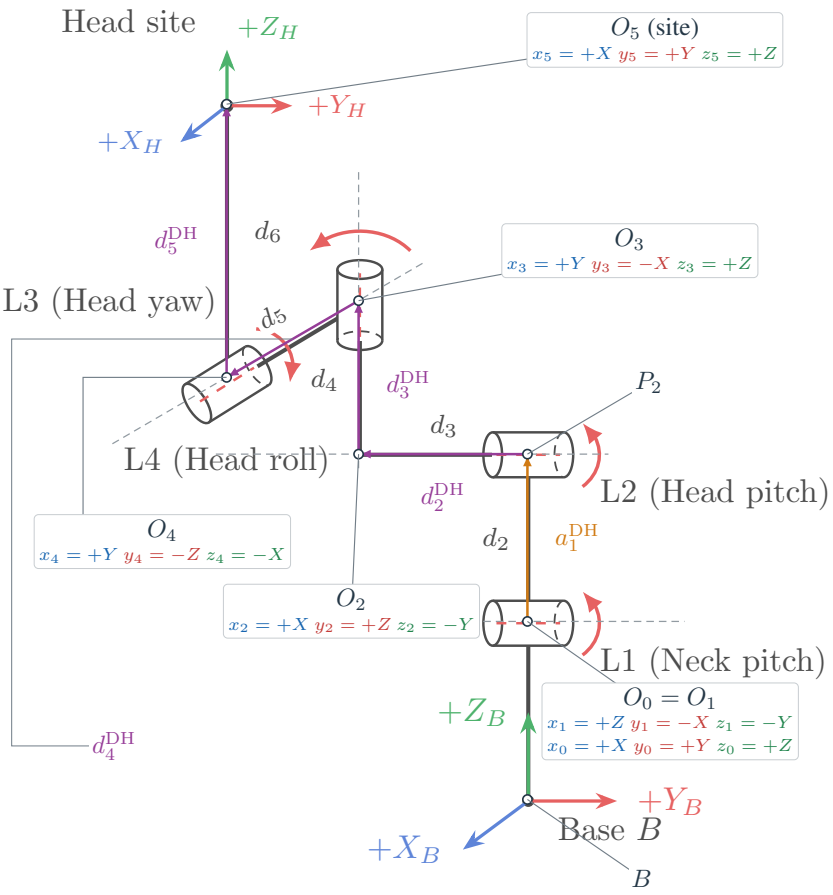


Figure 4: Head: annotated assignment Fig. 4. O_i: D-H origins; P_i: intermediate points. Orange: a_{i-1}; purple: d_i. Boxes give positive XYZ directions in B.

Table 3: Head: D–H parameters (m, rad) and zero-pose axes in frame B.

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	$\pi/2$	0	0	$q_1 + \pi/2$
2	0	0.066	0.01905	$q_2 - \pi/2$
3	$-\pi/2$	0	0.03785	$q_3 + \pi/2$
4	$-\pi/2$	0	-0.005	q_4
5*	$\pi/2$	0	0.04245	$-\pi/2$

The 92.021 mm base-to-neck offset is already in $\mathbf{T}_{0,H}$; do not add it again. * Fixed head site.

i	$+x_i$	$+y_i$	$+z_i$
0	$+X$	$+Y$	$+Z$
1	$+Z$	$-X$	$-Y$
2	$+X$	$+Z$	$-Y$
3	$+Y$	$-X$	$+Z$
4	$+Y$	$-Z$	$-X$
5*	$+X$	$+Y$	$+Z$

X, Y, Z are base-frame unit axes. $O_0 = O_1$, but their axis directions differ.

2.5 Reading the annotated figures

The original assignment drawings from Figs. 2–4 are annotated with the D-H origins, intermediate points and translation components used by the tables. The original dimension labels and base/site axes are retained.

The original labels L_i identify the successive joints, denoted J_i here. Dots O_i mark selected D-H origins on their axes, not cylinder centres. The final origin is the site S, a measurement frame rather than an extra joint.

D-H origins can slide along a joint axis when choosing common normals. They need not coincide with the centres of the original joint cylinders. For example, the head-pitch origin is shifted along its axis by $19.05\,\mathrm{mm}$; this does not introduce a new joint or change the mechanism. The coloured boxes specify all three positive axis directions at each origin, expressed in the base frame. Every displayed frame is right-handed: $x_i \times y_i = z_i$.

Blue/red/green denote positive $x/y/z$. The modified D-H joint axis is z_i; the physical positive axis is $s_i z_i$. For example, the left hip-pitch axis is $-Y_B$ even though the selected z_3 is $+Y_B$. Arrow lengths are illustrative; the tables give the exact axis directions.

For each leg, $O_B = O_0$ denotes coincident origins, not equal frame orientations. For the head, $O_0 = O_1$ while their axes differ, and the initial fixed transform connects B to O_0. Joints J_3 (head yaw) and J_4 (head roll) are distinct even though their selected D-H origins are only $5\,\mathrm{mm}$ apart.

Each row follows $O_{i-1} \xrightarrow{a_{i-1}} P_i \xrightarrow{d_i} O_i$. Orange and purple identify these two components. The handout perspective is schematic; the accompanying tables give the exact signed lengths and axis directions in B.

2.6 Zero-pose check

Using the dimension labels in the assignment handout (converted to metres), the zero-angle positions are

$$\mathbf{p}_L(\mathbf{0}) = \begin{bmatrix} -d_{1x} + d_5 + d_{9x} \\ d_{1y} + d_6 - d_{9y} \\ -d_4 - d_7 - d_8 - d_{9z} \end{bmatrix}, \qquad \mathbf{p}_R(\mathbf{0}) = \begin{bmatrix} -d_{1x} + d_5 + d_{9x} \\ -d_{1y} - d_6 + d_{9y} \\ -d_4 - d_7 - d_8 - d_{9z} \end{bmatrix}.$$

For the head dimensions, $\mathbf{p}_H(\mathbf{0}) = \mathbf{p}_c + [d_5, -d_3, d_2 + d_4 + d_6]^\mathsf{T}$. The resulting positions match the XML model. All three site rotations are the identity matrix at zero pose.

Table 4: Zero-pose site positions in the base frame.

Site	x (m)	y (m)	z (m)
Left foot	-0.03456	0.08940	-0.1935991
Right foot	-0.03456	-0.09180	-0.1935991
Head	0.00600	0.00010	0.2363009

3 Inverse kinematics

3.1 Error function

For a target pose $(\mathbf{p}_t, \mathbf{R}_t)$ and the FK pose $(\mathbf{p}(\mathbf{q}), \mathbf{R}(\mathbf{q}))$, the position error is $\mathbf{e}_p = \mathbf{p}_t - \mathbf{p}(\mathbf{q})$. The orientation error $\mathbf{e}_R$ is the rotation vector of $\mathbf{R}_t\mathbf{R}(\mathbf{q})^\mathsf{T}$, calculated using `Rotation.as_rotvec()`. Its length is the smallest relative rotation angle; Euler angles are not subtracted.

Following the assignment and supplied template, the cost is

$$E(\mathbf{q}) = \|10\mathbf{e}_p\|^2 + \|\mathbf{e}_R\|^2. \tag{2}$$

The position scale is $10\ \mathrm{m}^{-1}$. Each trial cost uses the FK model without changing MuJoCo's state. The numerical settings follow the assignment template.

3.2 Solution steps

The solver starts from the selected chain's current joint angles and reads its joint limits from the model. Each iteration estimates the gradient with a central difference, using $h = 10^{-6}\ \mathrm{rad}$:

$$g_i = \frac{E(\mathbf{q} + h\mathbf{u}_i) - E(\mathbf{q} - h\mathbf{u}_i)}{2h}.$$

Here $\mathbf{u}_i$ selects joint i. The solver then takes a fixed step and clips each joint to its allowed range:

$$\mathbf{q}_{\mathrm{new}} = \mathrm{clip}(\mathbf{q} - 0.1\mathbf{g}, \mathbf{q}_{\min}, \mathbf{q}_{\max}). \tag{3}$$

It stops when $E < 10^{-8}$, or after 5,000 iterations. Since both cost terms are nonnegative, the cost threshold implies FK-model errors below $10^{-5}\ \mathrm{m}$ and $10^{-4}\ \mathrm{rad}$. Convergence is not guaranteed for every possible target, so the final joint angles are written to MuJoCo and the actual site pose is checked independently.

3.3 Test trajectories

The supplied trajectories and starting angles are retained. For phase $\phi \in [0, 1]$, the feet move forward and upward while keeping their starting orientations:

$$\Delta\mathbf{p}_L = 0.020\sin^2(\pi\phi)(\mathbf{e}_x + \mathbf{e}_z), \qquad \Delta\mathbf{p}_R = 0.020\cos^2(\pi\phi)(\mathbf{e}_x + \mathbf{e}_z).$$

The head follows a reachable arc about the neck anchor. With $\gamma = 0.1\sin^2(\pi\phi)$,

$$\mathbf{p}_{t,H} = \mathbf{p}_c + \mathbf{R}_y(-\gamma)(\mathbf{p}_{H,0} - \mathbf{p}_c), \qquad \mathbf{R}_{t,H} = \mathbf{R}_y(-\gamma)\mathbf{R}_{H,0}.$$

The initial joint angles are

$$\mathbf{q}_{L,0} = [0, 0, -0.4, 0.8, -0.4]^\mathsf{T}, \quad \mathbf{q}_{R,0} = [0, 0, 0.4, 0.8, -0.4]^\mathsf{T},$$
$$\mathbf{q}_{H,0} = [0.2, -0.4, 0, 0]^\mathsf{T}.$$

The base is held at $0.19\ \mathrm{m}$. Each new target starts from the previous solution, including at cycle boundaries. The test runs five cycles of 101 phases for each of the three chains.

3.4 Robot pose before and after IK

The pictures below use the same model, starting joint angles and solver as the trajectory test. Both views show the target at $\phi = 0.5$. The solved pose is obtained by following the first 51 phases $(0, 0.01, \ldots, 0.5)$, starting each target from the previous solution.

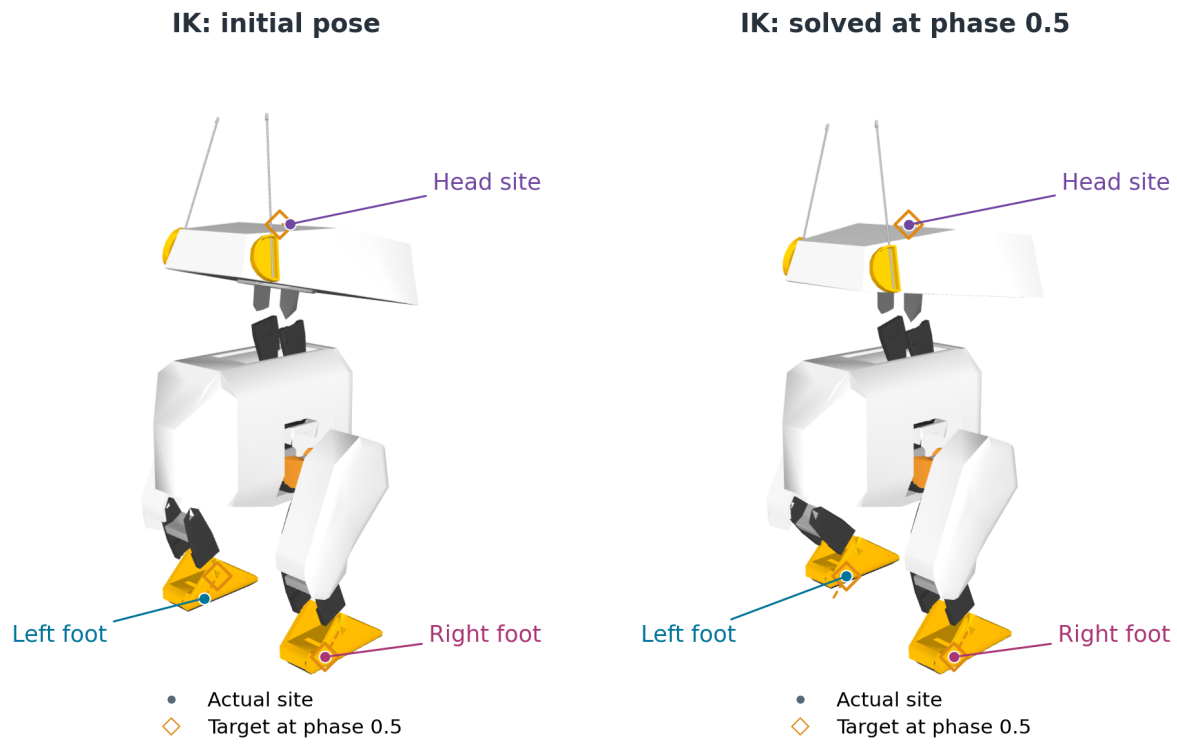


Figure 5: IK on the actual robot model: the joint angles change to bring the site positions to the target markers.

At this phase, the left foot moves $20\ \mathrm{mm}$ forward and $20\ \mathrm{mm}$ upward from its starting pose. The right-foot target returns to its starting pose, and the head target follows a $0.1\ \mathrm{rad}$ arc about the neck. The base remains fixed.

The visible markers show position only. The IK objective also matches orientation: the feet retain their initial orientation, while the head rotates with its target arc. Small remaining errors are checked numerically; the images are not used to judge the error tolerance. Coloured circles mark the actual sites; orange diamonds mark the mid-cycle target positions. Dashed orange lines trace the first half-cycle.

4 Results and checks

4.1 Comparison method

MuJoCo world-frame poses are converted to the base frame using ${}^{B}\mathbf{T}_{S} = ({}^{W}\mathbf{T}_{B})^{-1}{}^{W}\mathbf{T}_{S}$. Position error is the distance between the two site origins. Orientation error is the angle of the relative rotation. Both errors are checked.

FK uses 100 random joint configurations per chain, sampled within joint limits with seed zero. IK checks all 1,515 chain poses along the test trajectories. Environment details are saved with the results in `environment.json`. This report uses the saved measurements without rerunning the simulation.

4.2 Maximum errors

The FK limits are 10^{-6} m for position and 10^{-5} rad for orientation. The IK limits are 10^{-5} m and 10^{-4} rad. Every measured error is finite and strictly below its limit.

Table 5: Measured FK errors over 300 random configurations. Maxima are computed independently for each chain.

Chain	Max. position (m)	Max. orientation (rad)	Result
Left leg	8.455733×10^{-15}	2.283510×10^{-14}	PASS
Right leg	4.429498×10^{-14}	2.375025×10^{-14}	PASS
Head	6.572043×10^{-15}	1.032631×10^{-13}	PASS

Table 6: Measured IK errors over 1,515 trajectory poses. Maxima are computed independently for each chain.

Chain	Max. position (m)	Max. orientation (rad)	Result
Left leg	9.847942×10^{-6}	6.846493×10^{-5}	PASS
Right leg	9.844913×10^{-6}	7.038121×10^{-5}	PASS
Head	8.393151×10^{-6}	6.773237×10^{-5}	PASS

The largest IK position error is 9.847942×10^{-6} m (about 0.00985 mm), compared with the 0.01 mm limit. The largest IK orientation error is 7.038121×10^{-5} rad. The very small FK errors show agreement with the XML geometry, rather than the accuracy of a physical robot.

4.3 Additional checks

All twelve regression tests pass. Six check the kinematics: zero poses, random poses with a moved base, the cost function, unchanged targets, nearby reachable targets and joint limits. They also check that solving one chain leaves the base and other chains unchanged. The other six check source-file verification and Windows report building.

MuJoCo's `mj_forward` updates the site poses without dynamic stepping during the comparisons. These tests therefore check kinematics, not contact stability, physical calibration or real-robot tracking.

4.4 Error over the complete trajectory

Figure 6 shows every recorded IK sample. Errors are divided by their limits, so values below one pass. The five cycles are separated by vertical lines. No samples are removed or smoothed.

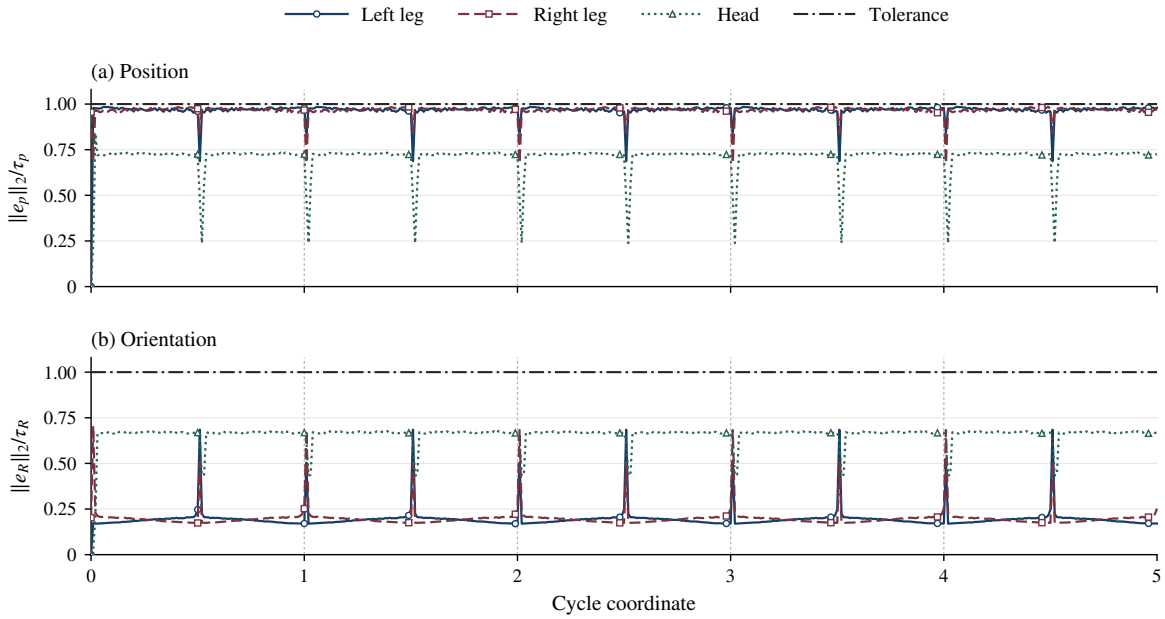


Figure 6: Position and orientation errors across the five IK cycles.

The leg position errors remain close to the limit because the solver stops once its weighted cost is small enough. It is not expected to drive every target to machine precision. All three chains remain below both acceptance limits.

5 Summary

The three D-H models reproduce the foot and head poses in the supplied MuJoCo model. The fixed-step IK solver follows the required trajectories within the specified position and orientation limits. The main details to handle carefully are the joint signs, the final fixed site row and the conversion from world coordinates to base coordinates.